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Pledgeability and Asset Prices: Evidence from the Chinese Corporate Bond Markets

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1 Big picture: what is identified and why it matters

Core object. The paper estimates the *value of pledgeability*, denoted λ , defined as the marginal (shadow) value of relaxing collateral constraints for marginal investors. In a collateralized funding system (repo), an asset that can be pledged more easily (lower haircut) delivers an extra service flow to constrained investors. This service flow is often called a *convenience yield* or *pledgeability premium*.

Empirical challenge. Pledgeability (haircuts, collateral eligibility) is usually endogenous to fundamentals and market conditions. A simple regression of yields on haircuts is therefore contaminated by omitted fundamentals and equilibrium feedback.

Identification strategy. China offers an unusual institutional design:

- The **same enterprise bond** can trade on two segmented markets: the exchange (EX) and the over-the-counter interbank market (IB).
- Repo rules differ across markets; exchange haircuts are set by a centralized depository (CSDC) via published *conversion rates*, while interbank haircuts are negotiated OTC.
- A sharp **policy shock** (Dec 8, 2014) made AA+/AA enterprise bonds ineligible as repo collateral on the exchange only, generating a large quasi-exogenous shift in exchange-market haircuts.

Main quantitative conclusion. When the exchange haircut increases from 0% to 100% (i.e., pledgeability collapses), bond yields increase by about **39 bps to 85 bps**. This range is interpreted as the magnitude of the collateral convenience yield embedded in corporate bond prices in China, and it implies a nontrivial shadow cost of capital for constrained investors.

2 Institutional background and measurement objects

2.1 Two markets and segmentation

- **Exchange market (EX):** centralized trading and repo via the exchange system. Exchange repo haircuts are standardized through *conversion rates* announced by the CSDC.
- **Interbank market (IB):** OTC market where repo haircuts are negotiated across counterparties.
- **Segmentation:** investor access differs (e.g., banks mainly on IB; retail on EX), and there are nontrivial frictions (e.g., settlement delays) that limit cross-market arbitrage.
- **Common marginal investors:** nonbank financial institutions (NBFIs) trade in both markets and are modeled as the relevant *common* marginal investors whose pricing kernel is shared across markets.

2.2 From conversion rates to haircuts

On the exchange, the CSDC sets a conversion rate CR (borrowable amount as a fraction of face value). Let FV be face value and P the market price. The exchange-market haircut is defined by:

$$(1 - \text{haircut}) \cdot P = CR \cdot FV \quad \Rightarrow \quad \text{haircut} = 1 - \frac{FV \cdot CR}{P}. \quad (1)$$

Interpretation: lower haircuts $\Rightarrow$ higher pledgeability $1 - h$ (more funding per unit pledged).

2.3 Exchange premium as a within-asset cross-market wedge

For dual-listed bonds traded on both markets, define the (yield) exchange premium:

$$EX \text{ premium}_{ijt} \equiv yield_{ijt}^{IB} - yield_{ijt}^{EX}, \quad (2)$$

where i indexes bonds, j the (possibly time-varying) rating category, and t time. A *negative EX premium* means exchange yields are higher than interbank yields (exchange prices are lower).

2.4 The December 8, 2014 policy shock

After market close on Dec 8, 2014, the exchange suspended repo eligibility for enterprise bonds rated below AAA. In this paper's setting:

- AA− bonds were already ineligible pre-shock (haircut $\approx 100\%$ on EX).
- AAA bonds remained eligible (low EX haircuts).
- AA+ and AA bonds experienced a large jump in EX haircuts (toward 100%).
- The shock is *exchange-market-specific*; IB repo rules did not mechanically change.

3 Economic model: pledgeability premium in bond prices

3.1 Collateral constraint pricing with pledgeability

Consider a one-period bond i (unit face value) with rating j and payoff $\tilde{Y}_{i,t+1}$. It trades in market $m \in \{EX, IB\}$ at price p_{ijt}^m and haircut h_{ijt}^m .

A marginal investor in market m faces a collateral constraint. The Euler equation can be written as:

$$p_{ijt}^m = \underbrace{\mathbb{E}_t \left[\tilde{M}_{t+1}^{I_m} \tilde{Y}_{i,t+1} \right]}_{\text{fundamental value}} + \underbrace{\lambda_t^{I_m} \cdot (1 - h_{ijt}^m)}_{\text{pledgeability premium}}. \quad (3)$$

Key objects:

- $\tilde{M}_{t+1}^{I_m}$: pricing kernel (SDF) of the marginal investor group I_m .
- $\lambda_t^{I_m}$: scaled Lagrange multiplier on the collateral constraint (shadow value of collateral services).
- $(1 - h)$: *pledgeability units* per unit face value.

3.2 Cross-market price wedge under common marginal investors

A crucial institutional feature is that NBFIs are marginal investors in *both* markets. Assume their pricing kernel is shared across markets and (within the event window) $\lambda_t^{NBFI} = \lambda$. Then differencing (3) across markets for the same bond yields:

$$p_{ijt}^{EX} - p_{ijt}^{IB} = \lambda (h_{ijt}^{IB} - h_{ijt}^{EX}). \quad (4)$$

Identification logic: fundamentals $\mathbb{E}_t[\tilde{M}\tilde{Y}]$ cancel within the same bond, so cross-market pricing differences isolate the wedge driven by pledgeability (and any other market-specific liquidity terms).

4 Event-study reduced form (diff-in-diff)

4.1 Treatment assignment

Define a treated rating group indicator for the post-shock window:

$$D_{jt} = \begin{cases} 1, & j \in \{AA^+, AA\} \text{ and } t > \text{Dec 8, 2014}, \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

4.2 Dynamic diff-in-diff in event time

Split the sample window into 14 subperiods of 28 calendar days, indexed by $k \in \{-6, \dots, 0, 1, \dots, 7\}$, with $k = 0$ the period immediately before the shock. Let D_{jt}^k indicate treated ratings in post period k . Then the event-time diff-in-diff specification is:

$$EX \text{ premium}_{ijt} = \sum_{k=-6}^7 d_k D_{jt}^k + a_i + b_j + c_t + X'_{it}e + u_{ijt}. \quad (6)$$

Role of (6): this checks the *parallel trend* assumption and shows dynamic persistence of the treatment effect.

5 2SLS estimation of the value of pledgeability

5.1 Allowing for other liquidity factors

To accommodate market-specific liquidity forces beyond pledgeability, augment (4) with residual terms:

$$p_{ijt}^{EX} - p_{ijt}^{IB} = \lambda (h_{ijt}^{IB} - h_{ijt}^{EX}) + \mu_{ijt}^{EX} - \mu_{ijt}^{IB}. \quad (7)$$

5.2 Structure on residuals and unobserved interbank haircuts

Two key empirical assumptions are imposed:

Residual decomposition.

$$\mu_{ijt}^m = \mu_{ijt} + \mu_i^m + \mu_j^m + \mu_t^m + \varepsilon_{ijt}^m, \quad (8)$$

so that rating-time market-specific residual variation is ruled out (the important potential violation is a market-specific flight-to-quality, discussed later).

Interbank haircuts vary only through common components. Since bond-level IB haircuts are not observed in the main dataset, assume:

$$h_{ijt}^{IB} = h_i^{IB} + h_j^{IB} + h_t^{IB}. \quad (9)$$

5.3 Fixed-effect representation and why IV is needed

Define $\Delta\mu_u \equiv \mu_u^{EX} - \mu_u^{IB}$ for $u \in \{i, j, t\}$. Combining (7)–(9) yields:

$$p_{ijt}^{EX} - p_{ijt}^{IB} = -\lambda \cdot h_{ijt}^{EX} + \underbrace{(\lambda h_i^{IB} + \Delta\mu_i)}_{\alpha_i} + \underbrace{(\lambda h_j^{IB} + \Delta\mu_j)}_{\alpha_j} + \underbrace{(\lambda h_t^{IB} + \Delta\mu_t)}_{\alpha_t}. \quad (10)$$

Thus, λ is identified from *rating-time variation* in exchange haircuts h_{ijt}^{EX} . Because h_{ijt}^{EX} may still co-move with residual liquidity or measurement error, the paper instruments haircut changes using the rating-based policy shock.

5.4 2SLS specification

First stage (haircut response to the shock).

$$haircut_{ijt} = \beta D_{jt} + \rho_i + \kappa_j + \eta_t + X'_{it}\gamma + v_{ijt}. \quad (11)$$

Second stage (exchange premium response to fitted haircuts).

$$EX \text{ premium}_{ijt} = \delta \widehat{haircut}_{ijt} + \alpha_i + \alpha_j + \alpha_t + X'_{it}\theta + \xi_{ijt}. \quad (12)$$

Mapping to the model. The coefficient of interest satisfies:

$$\delta = -\lambda.$$

Hence, when haircut increases from 0 to 100 (a full collapse in pledgeability), the implied yield effect is $|\delta|$ percentage points.

6 Arbitrage bounds and the direction of bias

Cross-market arbitrage is limited but not impossible. Let C denote an effective arbitrage bound arising from frictions (e.g., settlement delays). Then the exchange premium wedge is bounded:

$$p_{ijt}^{EX} - p_{ijt}^{IB} = \max \left\{ \min [\lambda(h_{ijt}^{IB} - h_{ijt}^{EX}), C], -C \right\}. \quad (13)$$

Intuition: when arbitrage is easier (smaller C), the observed price wedge compresses toward zero even if λ is large, biasing exchange-premium-based estimators downward.

7 Matched-AAA approach as an upper-bound design

7.1 Motivation

If exchange-premium-based estimates are biased downward (arbitrage compression and potential flight-to-quality), a complementary strategy is to estimate the treatment effect using the exchange market only, benchmarking treated AA+/AA bonds against *matched* AAA bonds with similar pre-event characteristics.

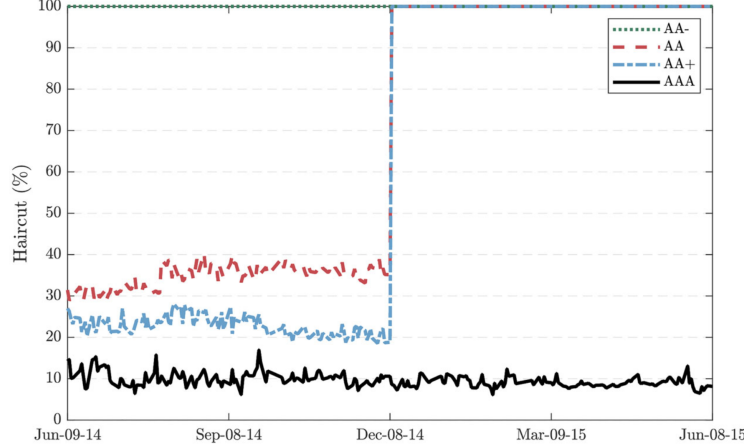


Figure 1. Average repo haircut on the exchange market. This figure plots the average daily haircut on the exchange market for dual-listed enterprise bonds in each of the four rating categories. The sample period is June 9, 2014 to June 8, 2015.

7.2 Matched-AAA pricing wedge

For a treated bond matched to a AAA bond (matched on pre-event haircut and credit spread), the exchange-market price wedge satisfies:

$$p_{treated,t}^{EX} - p_{matchedAAA,t}^{EX} = \lambda (h_{matchedAAA,t}^{EX} - h_{treated,t}^{EX}) + \mathbb{E}_t \left[\tilde{M}_{t+1}^{NBFI} (\tilde{Y}_{treated,t+1} - \tilde{Y}_{matchedAAA,t+1}) \right] + \underbrace{\mu_{treated,t}^{EX} - \mu_{matchedAAA,t}^{EX}}_{\text{liquidity residual}}. \quad (14)$$

Upper-bound logic: many plausible threats to identification (e.g., treated bonds having worse unobserved fundamentals) induce a *positive* correlation between the shock and the residual terms in (14), making the estimated effect larger in magnitude.

8 Interpreting the empirical results (tables and figures)

This section reads the empirical evidence in the order it appears in the paper. I focus on *what each exhibit is doing for identification* and *how to interpret its magnitudes*.

8.1 Figure 1: the policy shock is a large and rating-specific haircut jump

Figure 1 plots average exchange haircuts by rating over the sample window.

- Pre-shock: AAA haircuts are low and stable; AA+/AA haircuts are meaningfully higher; AA− is near 100%.
- Post-shock: AA+/AA haircuts jump essentially to 100% immediately, while AAA stays low.

Interpretation: the shock generates a powerful first-stage move in pledgeability that is sharply discontinuous in ratings.

8.2 Table I: sample coverage and why dual listing matters

Table I documents three layers of coverage:

- (a) **Dual-listed sample size:** 1,912 dual-listed enterprise bonds; 1,028 are in the simultaneous-trading sample.
- (b) **Coverage of enterprise bonds:** dual-listed bonds cover about 82% of enterprise bonds by count and about 78% by notional.
- (c) **Enterprise bonds within all corporate bonds:** enterprise bonds are an economically important subset (around one-quarter by notional, and a much larger share of trading volume for some ratings).

Interpretation: the dual-listed design provides within-asset cross-market variation, while the high coverage supports external relevance for the enterprise-bond segment.

Table I
Sample Coverage

This table reports the sample coverage by rating. Panel A presents the number of bonds for the dual-listed enterprise bond sample, the simultaneous trading sample, and the simultaneous trading sample with MCB only. Panel B presents the dual-listed enterprise bond sample coverage over all enterprise bonds. Panel C presents the enterprise bond sample coverage over all corporate bonds. Sample coverage measures in Panels B and C include number of bonds, notional RMB value, number of nonzero trading days, and RMB trading volume.

Panel A: Dual-Listed Sample and Simultaneous-Trading Sample					
	All	AAA	AA+	AA	AA-
$N_{dual-listed}$	1,912	234	578	981	119
$N_{simultaneous}$	1,028	83	318	536	91
$N_{meb_simultaneous}$	894	49	279	490	76
Panel B: Dual-Listed Sample Relative to All Enterprise Bonds					
	All	AAA	AA+	AA	AA-
Number of bonds	81.7%	60.5%	82.5%	87.8%	88.1%
Notional value	78.3%	59.2%	83.6%	88.5%	90.1%
Days with trades	92.1%	83.3%	92.2%	93.0%	97.2%
RMB trading volume	82.7%	55.1%	78.8%	90.9%	90.6%
Panel C: Enterprise Bonds Relative to All Corporate Bonds					
	All	AAA	AA+	AA	AA-
Number of bonds	28.0%	21.6%	38.8%	48.8%	5.5%
Notional value	26.5%	18.8%	37.6%	56.4%	5.5%
Days with trades	41.5%	25.5%	53.0%	57.9%	19.7%
RMB trading volume	26.7%	13.1%	29.8%	66.8%	4.6%

Table II
Summary Statistics

This table reports summary statistics for the simultaneous trading sample from June 9, 2014 to June 8, 2015. The table presents number of observations, mean, standard deviation, 10th percentile, median, and 90th percentile. Panel A presents summary statistics for key variables. Panel B presents summary statistics for exchange premia by rating. Panel C presents summary statistics for haircuts by rating.

Panel A: All Variables						
	N	Mean	SD	P10	Median	P90
EX premium	10,235	-0.04	0.48	-0.63	-0.02	0.50
EX premium _{pre}	5,069	0.07	0.40	-0.39	0.04	0.55
EX premium _{post}	5,166	-0.15	0.53	-0.76	-0.12	0.42
Haircut	10,235	68.64	38.01	15.77	100.00	100.00
Haircut _{pre}	5,069	42.32	32.60	8.12	30.90	100.00
Haircut _{post}	5,166	94.48	21.74	100.00	100.00	100.00
Conversion	10,235	33.24	40.37	0.00	0.00	88.00
Conversion _{pre}	5,069	61.22	34.79	0.00	73.00	97.00
Conversion _{post}	5,166	5.79	22.81	0.00	0.00	0.00
IB spread	10,235	2.41	0.79	1.42	2.44	3.40
EX spread	10,235	2.45	0.86	1.34	2.51	3.48
Matched spread	9,940	0.55	0.68	-0.15	0.47	1.38
Matched spread _{pre}	2,227	0.06	0.16	-0.13	0.04	0.27
Matched spread _{post}	7,713	0.69	0.71	-0.16	0.70	1.49
Matched spread _{AA+}	7,570	0.54	0.67	-0.14	0.46	1.37
Matched spread _{AA}	2,370	0.56	0.71	-0.16	0.48	1.43
$\Delta P^{high-low}$	10,235	0.44	1.44	-0.21	0.00	1.83
Maturity	10,235	5.10	1.61	2.97	5.26	6.72
Turnover	10,235	0.08	0.08	0.02	0.05	0.17
Market price	10,235	104.97	5.76	100.36	105.36	110.72
Volatility	10,235	0.02	0.02	0.00	0.01	0.04
CDB _{spot}	10,235	0.04	0.01	0.04	0.04	0.05
Term spread	10,235	0.01	0.00	0.00	0.00	0.01
GC001-SHIBOR	10,235	0.02	0.04	-0.00	0.01	0.06
Ret _{stock}	10,235	0.00	0.02	-0.01	0.00	0.02
Panel B: Exchange Premia by Rating (%)						
AAA	477	0.10	0.37	-0.37	0.03	0.59
AA+	3,077	0.01	0.48	-0.55	0.01	0.55
AA	5,162	-0.09	0.50	-0.71	-0.05	0.47
AA-	1,519	-0.02	0.45	-0.49	-0.01	0.47
Panel C: Haircuts by Rating (%)						
AAA	477	11.26	10.03	5.48	6.81	26.28
AA+	3,077	62.32	40.58	7.44	100.00	100.00
AA	5,162	68.49	35.46	29.81	100.00	100.00
AA-	1,519	100.00	0.00	100.00	100.00	100.00

8.3 Table II: summary statistics connect the shock to the wedge

Table II reports the main objects:

- **Exchange premium:** mean shifts from slightly positive pre-shock to negative post-shock, consistent with exchange yields rising relative to interbank yields for treated bonds.
- **Haircut:** average exchange haircuts increase substantially post-shock, with post-shock medians at 100% due to ineligibility.
- **By rating:** AAA haircuts are low; AA- haircuts are 100%; AA+/AA sit in between pre-shock but become effectively non-pledgeable post-shock.

Interpretation: the raw moments already align with the model (4): reduced pledgeability on EX implies lower EX prices and thus lower *EX premium*.

Table III
Determinants of Conversion Rates and Exchange Premia

This table reports regression results of dual-listed enterprise bonds' exchange market conversion rates (columns (1) and (2)) and exchange premia (columns (3) and (4)) on rating dummies and control variables. Age is the number of years for the issuer's first bond issuance, Nbond is the number of bonds issued by the issuer, and OTR is a dummy variable for on-the-run bond of the issuer. The sample period is June 9, 2014 to December 8, 2014. Heteroskedasticity-consistent standard errors clustered by week are reported in parentheses. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively. The standard error for AA- in column (1) is undefined because the conversion rates of AA- bonds are always zero.

	Conversion Rates		Exchange Premia	
	(1)	(2)	(3)	(4)
Haircut			-0.22*** (0.03)	-0.27*** (0.05)
Dummy _{AAA}	89.40*** (1.35)	-40.76 (25.23)		
Dummy _{AA+}	79.40*** (1.06)	-49.10* (25.54)		0.00 (0.05)
Dummy _{AA}	66.92*** (0.72)	-60.20** (25.29)		-0.04 (0.05)
Dummy _{AA-}	0.00 (-)	-124.94*** (24.76)		0.04 (0.08)
Market price		0.95*** (0.20)		0.02*** (0.00)
Volatility		-20.36 (23.80)		-0.10 (0.89)
MCB		-3.29** (1.39)		0.10** (0.04)
Age		0.19 (0.36)		-0.02* (0.01)
Nbond		-0.34*** (0.11)		0.01** (0.00)
OTR		-1.72** (0.83)		0.03 (0.02)
Maturity		1.66*** (0.32)		-0.03*** (0.01)
Turnover		-1.97 (4.22)		0.34*** (0.11)
Size		4.11*** (0.73)		0.01 (0.02)
Leverage		-16.24*** (3.56)		-0.09 (0.07)
Issuance		1.40* (0.74)		0.01 (0.01)
CDB _{spot}		291.43*** (96.20)		7.08 (4.36)
Term spread		539.71* (273.02)		-22.55* (12.39)
GC001-SHIBOR		-7.47 (5.80)		-0.22 (0.19)
Ret _{stock}		-5.23 (33.92)		-2.52* (1.27)
R ²	0.90	0.91	0.03	0.11
N	5,069	5,069	5,069	5,069

8.4 Table III: ratings largely determine exchange pledgeability, and haircuts price the wedge

Table III has two purposes:

- Conversion-rate determination:** rating dummies explain about 90% of conversion-rate variation. Adding bond/issuer controls raises R^2 only marginally.
- Exchange-premium correlation:** exchange premia are negatively correlated with exchange haircuts (higher haircut $\Rightarrow$ more negative exchange premium), consistent with pledgeability pricing.

Interpretation: exchange pledgeability is essentially *rating-driven*, strengthening the credibility of using rating-based policy variation as an instrument.

8.5 Table IV and Figure 2: market reactions isolate an exchange-specific pledgeability shock

Panel A of Table IV. On the policy-shock day, exchange credit spreads jump for the affected rating classes (AA+/AA), while interbank spreads do not display a mechanically similar jump. The implied IB–EX wedge moves substantially for treated bonds relative to controls.

Panel B of Table IV. Interbank haircuts (from a large institution) shift roughly in parallel across ratings, supporting assumption (9) that any time variation is largely common rather than rating-specific.

Table IV
Market Reactions to the Policy Shock and Other Events

This table reports the average market reactions to the policy shock and other events. Average one-day post-announcement changes in credit spreads are reported in Panel A. Average haircuts of an anonymous major financial institution on the interbank market six months prior to and after the policy shock are reported in Panel B. The policy shock was on December 8, 2014, the release of Doc. 43 was on October 2, 2014, and the five announcements were made on May 29, 2014, June 27, 2014, August 1, 2014, September 5, 2014, and November 3, 2014, respectively. Due to the lack of trades on September 30, 2014 before the National Holiday (October 1, 2014 to October 7, 2014), trades in the two-day window before the holiday are used to calculate the pre-Doc. 43 credit spreads. The post-pre credit spread difference between the interbank and the exchange markets for treatment and control groups is presented in the last two columns and estimated in a regression on $post_t \times 1_{IB}$ that includes the $post_t$ dummy, the interbank market indicator 1_{IB} , and $rating_j \times post_t$ fixed effects. Heteroskedasticity-robust standard errors are reported in parentheses. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Market Reactions by Market and Rating (bps)										
	EX				IB				IB–EX	
	AAA	AA+	AA	AA–	AAA	AA+	AA	AA–	AA+ & AA	AAA & AA–
Policy	–14.69	61.61***	37.64***	60.52***	–24.33	–7.97	–9.12	23.87	–55.31***	–31.23
Shock	(17.40)	(12.10)	(13.47)	(18.99)	(32.26)	(13.39)	(8.20)	(21.49)	(11.96)	(24.37)
Doc. 43	–17.97*	5.58	6.73	1.46	17.86	11.23	7.66	–11.29	4.24	8.49
	(10.51)	(9.07)	(12.25)	(11.75)	(19.33)	(12.35)	(9.46)	(23.93)	(10.99)	(17.98)
Five	–0.41	3.27	4.55	8.21	–4.42	8.23	4.86	–19.15	1.75	–11.19
Blacklists	(7.35)	(4.57)	(5.05)	(8.67)	(11.58)	(6.51)	(3.60)	(23.75)	(4.89)	(12.16)

Panel B: Haircuts on the Interbank Market (%)				
Sample period	AAA	AA+	AA	AA–
June 9, 2014 to December 8, 2014	8.38	12.93	32.03	35.66
	(0.56)	(0.96)	(1.53)	(7.01)
December 9, 2014 to June 8, 2015	13.76	14.38	31.23	37.20
	(0.44)	(1.25)	(1.28)	(8.89)
November 9, 2014 to December 8, 2014	7.41	11.44	28.85	33.64
	(0.85)	(1.87)	(3.12)	(14.11)
December 9, 2014 to January 8, 2015	17.24	16.53	32.14	37.18
	(1.10)	(2.24)	(2.88)	(22.37)

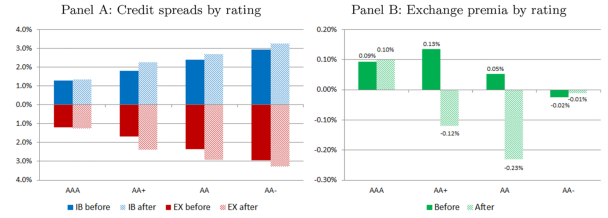


Figure 2. Exchange premia six months before and after the December 8, 2014 event. This figure plots the average credit spreads for each of the four rating categories on the interbank market and the exchange market (Panel A) and the average exchange premia (Panel B).

Figure 2.

- Panel A shows that post-shock, exchange spreads rise (especially for treated ratings).
- Panel B shows exchange premia become more negative for AA+/AA post-shock.

Interpretation: the combination of these exhibits supports the interpretation that the policy shock is primarily a pledgeability shock on the exchange market, not a fundamental shock common to both markets.

8.6 Figure 3: dynamic diff-in-diff confirms parallel pre-trends

Figure 3 plots d_k from (6) across event-time bins.

- Pre-event coefficients are statistically indistinguishable from zero: evidence for parallel trends.
- Post-event coefficients are negative and persistent: treated bonds exhibit a sustained decline in *EX premium*.

Interpretation: the reduced-form evidence is consistent with a persistent convenience-yield loss for treated bonds.

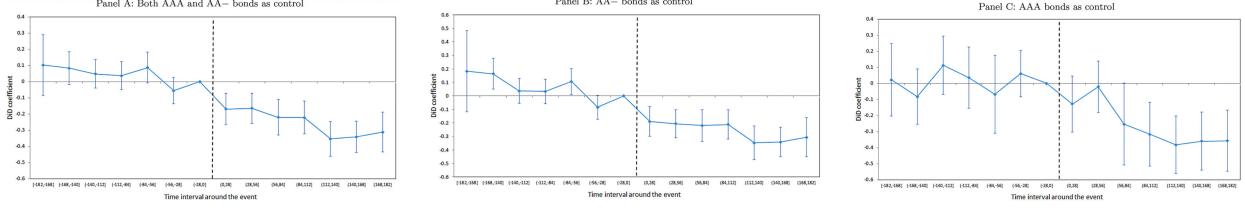


Figure 3. Diff-in-diff estimation of exchange premia.

8.7 Table V: baseline 2SLS estimate of λ (exchange premium approach)

Table V
IV Estimation

This table reports the results of IV regressions using the simultaneous trading sample. Panels A and B present results for the first- and second-stage regressions. Columns (1) and (2) present results using the full sample, without and with control variables, respectively. Column (3) presents results using a subsample of AA+, AA, and AA- bonds. Column (4) presents results using a subsample of AA+, AA, and AAA bonds. Column (5) presents results using a subsample of AA+, AAA, and AA- bonds. Column (6) presents results using a subsample of AA, AAA, and AA- bonds. The sample period is June 9, 2014 to June 8, 2015. Heteroskedasticity-consistent standard errors clustered by week are reported in parentheses. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: First Stage						
Dependent:	Full		Exclude AAA	Exclude AA-	Exclude AA	Exclude AA+
<u>Haircut</u>	(1)	(2)	(3)	(4)	(5)	(6)
<u>Shock</u>	68.00*** (0.64)	68.20*** (0.65)	68.27*** (0.67)	67.83*** (0.64)	74.82*** (0.88)	63.67*** (0.79)
Controls	—	✓	✓	✓	✓	✓
Bond FE	—	✓	✓	✓	✓	✓
Rating FE	✓	✓	✓	✓	✓	✓
Week FE	✓	✓	✓	✓	✓	✓
R ²	0.85	0.95	0.95	0.95	0.97	0.96
N	10,235	10,070	9,615	8,550	4,993	7,039
Panel B: Second Stage						
Dependent:	Full		Exclude AAA	Exclude AA-	Exclude AA	Exclude AA+
<u>EX Premia</u>	(1)	(2)	(3)	(4)	(5)	(6)
<u>Haircut</u>	-0.39*** (0.05)	-0.39*** (0.05)	-0.40*** (0.05)	-0.33*** (0.10)	-0.38*** (0.05)	-0.40*** (0.05)
Maturity	—	2.75*** (0.80)	2.84*** (0.82)	3.06*** (0.86)	3.30*** (1.02)	1.85* (0.93)
Turnover	—	0.12 (0.09)	0.10 (0.09)	0.13 (0.10)	0.23 (0.14)	0.09 (0.10)
Market price	—	-0.00 (0.00)	-0.00 (0.00)	-0.00 (0.00)	-0.01 (0.00)	-0.00 (0.00)
Volatility	—	0.03 (1.04)	-0.07 (1.06)	0.20 (1.13)	-0.65 (1.71)	0.24 (0.77)
CDB _{spot}	—	-28.15** (11.05)	-30.47*** (11.36)	-20.06* (11.36)	-34.60** (12.94)	-27.66* (15.22)
Term spread	—	27.95* (15.85)	28.48* (15.51)	28.01 (17.08)	36.52** (17.47)	23.77 (18.22)
GC001-SHIBOR	—	-0.15 (0.15)	-0.15 (0.15)	-0.07 (0.14)	-0.10 (0.19)	-0.27** (0.13)
Ret _{stock}	—	0.46 (0.43)	0.43 (0.43)	0.48 (0.49)	0.41 (0.42)	0.42 (0.47)
Bond FE	—	✓	✓	✓	✓	✓
Rating FE	✓	✓	✓	✓	✓	✓
Week FE	✓	✓	✓	✓	✓	✓
R ²	0.12	0.48	0.47	0.49	0.41	0.53
N	10,235	10,070	9,615	8,550	4,993	7,039

Table V contains the central causal estimate.

First stage (Panel A). The policy shock increases exchange haircuts by roughly 68–75 percentage points depending on the control group, with very high R^2 values. This confirms a strong instrument.

Second stage (Panel B). The estimated coefficient on *haircut* is about -0.39 to -0.40 (in percentage points).

- **Economic meaning:** raising haircut from 0% to 100% increases bond yields by about 39–40 bps.

Table VI
IV Estimation: Additional Results

This table reports additional results for the IV regressions. Panels A and B present results for the first- and second-stage regressions. Column (1, MCB) presents results using the MCBs only. Column (2, Maturity^{long}) presents results using a subsample of bonds for which the time-to-maturity as of the day of trade is above median. Column (3, Excl. Mth 1) presents results using the subsample without the first post-event month. Column (4, 2SWLS) presents results using two-stage weighted least squares, where the weight is equal to the inverse of the number of observations for each bond. Column (5, Continuous) presents results using $(1 - \text{haircut}^{\text{pre}})$ as the shock size for AA+ and AA bonds, where $\text{haircut}^{\text{pre}}$ is the average haircut for bond i rating j before the policy shock. Heteroskedasticity-consistent standard errors clustered by week are reported in parentheses. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: First Stage					
Dependent:	MCB	Maturity ^{long}	Excl. Mth 1	2SWLS	Continuous
Haircut	(1)	(2)	(3)	(4)	(5)
Shock	68.34*** (0.75)	69.65*** (0.95)	68.26*** (0.70)	67.49*** (0.76)	99.65*** (0.50)
Controls	✓	✓	✓	✓	✓
Bond FE	✓	✓	✓	✓	✓
Rating FE	✓	✓	✓	✓	✓
Week FE	✓	✓	✓	✓	✓
R ²	0.95	0.97	0.96	0.97	0.99
N	8,513	4,995	9,132	10,070	10,070

Panel B: Second Stage					
Dependent:	MCB	Maturity ^{long}	Excl. Mth 1	2SWLS	Continuous
EX Premia	(1)	(2)	(3)	(4)	(5)
Haircut	-0.34*** (0.05)	-0.46*** (0.06)	-0.43*** (0.04)	-0.41*** (0.06)	-0.34*** (0.05)
Controls	✓	✓	✓	✓	✓
Bond FE	✓	✓	✓	✓	✓
Rating FE	✓	✓	✓	✓	✓
Week FE	✓	✓	✓	✓	✓
R ²	0.47	0.58	0.51	0.57	0.48
N	8,513	4,995	9,132	10,070	10,070

Table VII
IV Estimation: Impacts on Liquidity

This table reports second-stage results of IV regressions using the difference in the price range between the exchange and interbank markets as the dependent variable. The price range in percentage is defined as the daily high minus daily low divided by the average of the two. Columns (1) and (2) present results using full sample, without and with control variables, respectively. Column (3) presents results using a subsample of AA+, AA, and AA− bonds. Column (4) presents results using a subsample of AA+, AA, and AAA bonds. Column (5) presents results using a subsample of AA+, AAA, and AA− bonds. Column (6) presents results using a subsample of AA, AAA, and AA− bonds. The sample period is June 9, 2014 to June 8, 2015. Heteroskedasticity-consistent standard errors clustered by week are reported in parentheses. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: First Stage						
Dependent:	Full		Exclude AAA	Exclude AA−	Exclude AA	Exclude AA+
Haircut	(1)	(2)	(3)	(4)	(5)	(6)
Shock	68.00*** (0.64)	68.23*** (0.66)	68.28*** (0.68)	67.97*** (0.67)	74.87*** (0.88)	63.68*** (0.79)
Controls	—	✓	✓	✓	✓	✓
Bond FE	—	✓	✓	✓	✓	✓
Rating FE	✓	✓	✓	✓	✓	✓
Week FE	✓	✓	✓	✓	✓	✓
R ²	0.85	0.95	0.95	0.95	0.97	0.96
N	10,235	10,070	9,615	8,550	4,993	7,039
Panel B: Second Stage						
Dependent:	Full		Exclude AAA	Exclude AA−	Exclude AA	Exclude AA+
$\Delta P^{high-low}$	(1)	(2)	(3)	(4)	(5)	(6)
Haircut	0.48*** (0.14)	0.41*** (0.14)	0.43** (0.16)	0.35* (0.20)	0.41** (0.17)	0.42*** (0.15)
Maturity	—	−10.59*** (3.59)	−10.31*** (3.64)	−11.65*** (3.98)	−10.39** (4.47)	−10.02*** (3.70)
Turnover	—	−2.50*** (0.28)	−2.50*** (0.29)	−2.32*** (0.31)	−3.05*** (0.46)	−2.41*** (0.34)
Market price	—	−0.01 (0.00)	−0.01 (0.00)	−0.01 (0.00)	−0.01 (0.01)	−0.00 (0.01)
CDB _{spot}	—	58.98 (45.94)	56.03 (46.35)	43.49 (60.70)	68.74 (48.97)	81.36* (48.18)
Term spread	—	19.09 (38.51)	25.52 (39.13)	25.91 (46.01)	−50.67 (46.45)	40.13 (41.31)
GC001−SHIBOR	—	−0.63 (0.79)	−0.51 (0.80)	−0.61 (0.87)	−1.70 (1.22)	0.13 (0.53)
Ret _{stock}	—	−0.58 (1.22)	−0.69 (1.25)	−0.71 (1.35)	−1.82 (2.39)	0.72 (0.74)
Bond FE	—	✓	✓	✓	✓	✓
Rating FE	✓	✓	✓	✓	✓	✓
Week FE	✓	✓	✓	✓	✓	✓
R ²	0.08	0.26	0.25	0.27	0.25	0.27
N	10,235	10,070	9,615	8,550	4,993	7,039

- **Model mapping:** since $\delta = -\lambda$, this implies $\lambda \approx 0.39$ (in annualized yield percentage points per unit haircut change).

8.8 Table VI: robustness and economic heterogeneity

Table VI demonstrates that the estimate is not driven by a single modeling choice:

- **MCB-only subsample:** similar magnitude, suggesting robustness within the dominant enterprise-bond subtype.
- **Long-maturity bonds:** larger effect (≈ 46 bps), consistent with the intuition that long-maturity/less liquid assets are more sensitive to collateral services.
- **Excluding the first post-event month:** the estimate remains sizable, reducing concerns about short-run transitional liquidity effects.
- **Alternative weighting (2SWLS) and continuous shock size:** the estimate remains in the same ballpark.

8.9 Table VII: pledgeability affects liquidity (price-range evidence)

Table VII uses the difference in daily price ranges (high minus low relative to midpoint) between EX and IB as the dependent variable. The 2SLS coefficient on *haircut* is positive and significant.

Interpretation: a negative pledgeability shock worsens relative liquidity/volatility on the exchange market, supporting a channel where collateral eligibility affects not only levels of yields but also trading conditions.

8.10 Figure 4 and Table VIII: matched-AAA approach provides an upper bound

Figure 4.

- Panel A shows that treated-minus-matched haircuts are approximately zero pre-event and jump sharply post-event.
- Panel B shows that treated-minus-matched exchange credit spreads rise post-event, mirroring the haircut jump.

Table V
IV Estimation

This table reports the results of IV regressions using the simultaneous trading sample. Panels A and B present results for the first- and second-stage regressions. Columns (1) and (2) present results using the full sample, without and with control variables, respectively. Column (3) presents results using a subsample of AA+, AA, and AA- bonds. Column (4) presents results using a subsample of AA+, AA, and AAA bonds. Column (5) presents results using a subsample of AA+, AAA, and AA- bonds. Column (6) presents results using a subsample of AA, AAA, and AA- bonds. The sample period is June 9, 2014 to June 8, 2015. Heteroskedasticity-consistent standard errors clustered by week are reported in parentheses. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: First Stage						
Dependent:	Full		Exclude AAA	Exclude AA-	Exclude AA	Exclude AA+
	(1)	(2)	(3)	(4)	(5)	(6)
<u>Haircut</u>						
Shock	68.00*** (0.64)	68.20*** (0.65)	68.27*** (0.67)	67.83*** (0.64)	74.82*** (0.88)	63.67*** (0.79)
Controls	—	✓	✓	✓	✓	✓
Bond FE	—	✓	✓	✓	✓	✓
Rating FE	✓	✓	✓	✓	✓	✓
Week FE	✓	✓	✓	✓	✓	✓
R ²	0.85	0.95	0.95	0.95	0.97	0.96
N	10,235	10,070	9,615	8,550	4,993	7,039
Panel B: Second Stage						
Dependent:	Full		Exclude AAA	Exclude AA-	Exclude AA	Exclude AA+
	(1)	(2)	(3)	(4)	(5)	(6)
<u>EX Premia</u>						
<u>Haircut</u>	-0.39*** (0.05)	-0.39*** (0.05)	-0.40*** (0.05)	-0.33*** (0.10)	-0.38*** (0.05)	-0.40*** (0.05)
Maturity		2.75*** (0.80)	2.84*** (0.82)	3.06*** (0.86)	3.30*** (1.02)	1.85* (0.93)
Turnover		0.12 (0.09)	0.10 (0.09)	0.13 (0.10)	0.23 (0.14)	0.09 (0.10)
Market price		-0.00 (0.00)	-0.00 (0.00)	-0.00 (0.00)	-0.01 (0.00)	-0.00 (0.00)
Volatility		0.03 (1.04)	-0.07 (1.06)	0.20 (1.13)	-0.65 (1.71)	0.24 (0.77)
CDB _{spot}		-28.15** (11.05)	-30.47*** (11.36)	-20.06* (11.36)	-34.60** (12.94)	-27.66* (15.22)
Term spread		27.95* (15.85)	28.48* (15.51)	28.01 (17.08)	36.52** (17.47)	23.77 (18.22)
GC001-SHIBOR		-0.15 (0.15)	-0.15 (0.15)	-0.07 (0.14)	-0.10 (0.19)	-0.27* (0.13)
Ret _{stock}		0.46 (0.43)	0.43 (0.43)	0.48 (0.49)	0.41 (0.42)	0.42 (0.47)
Bond FE	—	✓	✓	✓	✓	✓
Rating FE	✓	✓	✓	✓	✓	✓
Week FE	✓	✓	✓	✓	✓	✓
R ²	0.12	0.48	0.47	0.49	0.41	0.53
N	10,235	10,070	9,615	8,550	4,993	7,039

Table VI
IV Estimation: Additional Results

This table reports additional results for the IV regressions. Panels A and B present results for the first- and second-stage regressions. **Column (1, MCB) presents results using the MCBs only. Column (2, Maturity^{long}) presents results using a subsample of bonds for which the time-to-maturity as of the day of trade is above median.** Column (3, Excl. Mth 1) presents results using the subsample without the first post-event month. Column (4, 2SWLS) presents results using two-stage weighted least squares, where the weight is equal to the inverse of the number of observations for each bond. Column (5, Continuous) presents results using (1-haircut^{pre}) as the shock size for AA+ and AA bonds, where haircut^{pre} is the average haircut for bond *i* rating *j* before the policy shock. Heteroskedasticity-consistent standard errors clustered by week are reported in parentheses. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: First Stage					
Dependent:	MCB	Maturity ^{long}	Excl. Mth 1	2SWLS	Continuous
	(1)	(2)	(3)	(4)	(5)
<u>Haircut</u>					
Shock	68.34*** (0.75)	69.65*** (0.95)	68.26*** (0.70)	67.49*** (0.76)	99.65*** (0.50)
Controls	✓	✓	✓	✓	✓
Bond FE	✓	✓	✓	✓	✓
Rating FE	✓	✓	✓	✓	✓
Week FE	✓	✓	✓	✓	✓
R ²	0.95	0.97	0.96	0.97	0.99
N	8,513	4,995	9,132	10,070	10,070
Panel B: Second Stage					
Dependent:	MCB	Maturity ^{long}	Excl. Mth 1	2SWLS	Continuous
	(1)	(2)	(3)	(4)	(5)
<u>EX Premia</u>					
<u>Haircut</u>	-0.34*** (0.05)	-0.46*** (0.06)	-0.43*** (0.04)	-0.41*** (0.06)	-0.34*** (0.05)
Controls	✓	✓	✓	✓	✓
Bond FE	✓	✓	✓	✓	✓
Rating FE	✓	✓	✓	✓	✓
Week FE	✓	✓	✓	✓	✓
R ²	0.47	0.58	0.51	0.57	0.48
N	8,513	4,995	9,132	10,070	10,070

Table VIII. The 2SLS estimate using matched AAA bonds produces a larger magnitude: *haircut* coefficients around -0.74 to -0.85 . **Interpretation:** a 0% to 100% haircut increase implies a yield effect of roughly 74–85 bps. Because residual fundamental/liquidity differences can push estimates upward in this design, the paper interprets this as an *upper bound* for λ in this setting.

Table VIII
IV Estimation Using Matched AAA Bonds as Benchmark

This table reports results of IV regressions using the matched AAA bonds as a benchmark. The dependent variable is the credit spread between the matched AAA bonds and that of AA+/AA dual-listed enterprise bonds, where the matching criteria include the credit spread and the haircut before December 8, 2014. Panels A and B present results for the first and second stages. The sample period is June 9, 2014 to June 8, 2015. Heteroskedasticity-consistent standard errors clustered by week are reported in parentheses. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: First Stage				
Dependent:	Full		AA+	AA
	(1)	(2)	(3)	(4)
<u>Haircut</u>				
Shock	86.17*** (0.90)	84.85*** (1.23)	86.82*** (0.87)	77.67*** (1.35)
Controls	—	✓	✓	✓
Bond FE	—	✓	✓	✓
Rating FE	✓	✓	✓	✓
R^2	0.98	0.99	0.99	0.98
N	9,940	9,897	7,548	2,349
Panel B: Second Stage				
Dependent:	Full		AA+	AA
	(1)	(2)	(3)	(4)
<u>Spread^{matched-AAA}</u>				
<u>Haircut</u>	-0.74*** (0.02)	-0.85*** (0.04)	-0.84*** (0.04)	-0.84*** (0.10)
Maturity		0.03 (0.11)	0.07 (0.10)	-0.09 (0.20)
Turnover		2.22* (1.29)	1.23 (1.06)	5.94* (2.98)
Market price		-0.00 (0.00)	-0.00 (0.00)	0.01 (0.01)
Volatility		0.12 (0.99)	-1.03 (1.34)	2.19* (1.29)
CDB _{spot}		-10.28** (4.62)	-10.32** (4.15)	-7.96 (9.08)
Term spread		-0.91 (5.55)	-3.54 (4.74)	5.72 (10.30)
GC001-SHIBOR		-0.17 (0.30)	-0.12 (0.25)	-0.43 (0.54)
Ret _{stock}		0.77 (0.64)	1.09 (0.63)	0.11 (0.88)
Bond FE	—	✓	✓	✓
Rating FE	✓	✓	✓	✓
R^2	0.15	0.55	0.56	0.54
N	9,940	9,897	7,548	2,349

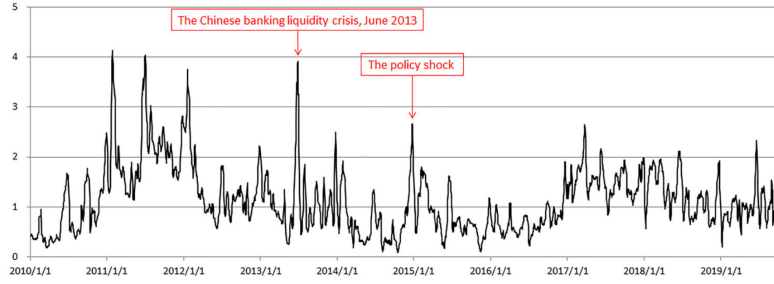


Figure 5. Spread between the interbank market repo rate and the CDB bond yield. This figure plots the daily spread in percentage between the one-month interbank market repo rate for all financial institutions and the CDB bond yield calculated from CDB bonds with one-month maturity. Two events, the CDSC policy shock on December 8, 2014 studied by this paper and the Chinese banking liquidity crisis during June 2013 analyzed in Hachem and Song (2021), are indicated. The sample period is January 1, 2010 to October 31, 2019. (Color figure can be viewed at wileyonlinelibrary.com)

8.11 Figure 5: why λ can be time-varying

Figure 5 plots a macro funding-stress proxy (interbank repo rate minus CDB yield) over a longer sample (2010–2019), marking the 2014 policy event and the June 2013 liquidity crisis. **Interpretation:** λ is conceptually a shadow price of collateral constraints and should be higher during funding stress. The paper’s event window assumes approximate constancy of λ , while Figure 5 motivates extensions with time-varying λ_t .

9 Short Summary

1. **Theoretical structure:** price = fundamental value + $\lambda \times$ pledgeability units.
2. **Within-asset cross-market design:** same bond trades on two markets, so fundamentals cancel in price differences.

3. **Instrument:** a rating-based policy shock that changes exchange repo eligibility but not interbank rules.
4. **Causal estimate:** losing pledgeability (haircut 0→100) raises yields by about 39–85 bps.
5. **Methodological lesson:** combine market segmentation, quasi-exogenous institutional shocks, and within-asset comparisons to identify collateral convenience yields.

10 Conclusion

10.1 Summary

This paper concludes that **asset prices depend on pledgeability in addition to fundamentals**, and it provides **causal evidence** on this relationship using a unique institutional setting in China.

The empirical environment delivers an unusually clean identification design:

- The same corporate bond (dual-listed enterprise bond) can trade in two *segmented* markets: the **exchange market** and the **OTC interbank market**.
- Repo (collateralized borrowing) rules differ sharply across the two markets. In the exchange market, repo eligibility and haircuts are set administratively (largely by rating); in the interbank market, repo terms are negotiated bilaterally.
- Nonbank financial institutions (NBFIs) participate in both markets, which makes it plausible that the *same* marginal pricing kernel applies to the same bond across venues, so cross-market price differences can isolate nonfundamental components such as pledgeability.

Exploiting a rating-based policy shock on December 8, 2014 that **reduced repo eligibility (raised haircuts) on the exchange market only** for the treated rating classes (notably AA+ and AA), the paper finds that when a bond’s haircut increases from 0% to 100%, its yield (credit spread) increases by about **39 bps to 85 bps**. These effects are interpreted as estimates (or bounds) on the **value of pledgeability**—that is, the shadow value of relaxing collateral constraints.

Finally, mapping these estimates into a collateral-constraint framework implies that the estimated pledgeability value corresponds to a **shadow cost of capital** for Chinese NBFIs on the order of roughly **1.1% to 2.4%** per year.

10.2 Key quantitative conclusions

Let $h \in [0, 1]$ denote the repo haircut (a higher h means *less* pledgeable collateral). The paper’s headline numbers can be summarized as:

- (i) **Main IV estimate (exchange-premium design):** a 100 percentage-point increase in haircut ($\Delta h = 1$) raises yields by about **39 bps**.
- (ii) **Alternative IV estimate (matched-AAA benchmark on exchange):** a 100 percentage-point increase in haircut raises yields by about **85 bps**.
- (iii) **Economic magnitude:** interpreting these yield effects as the per-unit value of pledgeability implies a nontrivial **shadow cost of capital** (roughly **1.1%–2.4%** annually) in China’s financial sector.

10.3 Why does pledgeability move bond yields?

The core intuition is a **collateral (funding) channel**:

1) Pledgeability creates a convenience yield. If investors face a funding/collateral constraint, then holding a bond that can be pledged with a low haircut reduces the amount of scarce internal capital needed to finance positions. That financing benefit is a **convenience yield** embedded in the bond price: more pledgeable bonds trade at *higher* prices (lower yields), even if cash flows are identical.

2) A simple reduced-form relationship. A useful way to summarize the mechanism (consistent with collateral-constraint asset-pricing logic) is:

$$y = y^{\text{fund}} + \lambda h + (\text{other liquidity wedges}), \quad (1)$$

where:

- y is the bond yield (or credit spread),
- y^{fund} is the component pinned down by default risk and fundamentals,
- h is the haircut (inverse pledgeability),
- λ is the **shadow value of collateral / shadow cost of capital**.

Then a haircut shock implies:

$$\Delta y \approx \lambda \Delta h. \quad (2)$$

Because the policy shock effectively moves treated bonds from $h = 0$ to $h = 1$ on the exchange market, the observed yield increase is a direct estimate (or bound) on λ .

3) Why the Chinese setting identifies this cleanly. The same bond trades in two markets, and NBFIs operate in both. By comparing yields across venues (“exchange premium”) and exploiting a policy shock that changes pledgeability *only* on one venue and *only* for certain ratings, the design differences out many confounders: unobserved bond fundamentals, issuer-level shocks, and broad macro conditions that would otherwise make pledgeability endogenous.

10.4 Takeaway

A large, rating-based repo-eligibility shock in China shows that collateral value is priced: when a bond’s haircut jumps from 0% to 100% on the exchange market, its yield rises by roughly 39–85 bps, implying a sizeable shadow value of collateral and a meaningful shadow cost of capital for constrained financial intermediaries.